Nurun Nahar Fiha

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Research Interest

Natural Language Processing, Reliability, Truthfulness, and Evaluation of Large Language Models, Low-Resource and Multilingual NLP, Human-AI Interaction

Education

Jahangirnagar University **Savar, Bangladesh**
M.Sc. in Computer Science and Engineering *Jun 2024 – Jan 2026*

- **CGPA: 3.83/4.00 (Rank: 7/44)**
- **Notable Courses:** Natural Language Processing, Pattern Recognition, Machine Intelligence & Expert Systems

B.Sc. in Computer Science and Engineering *Feb 2019 – Jul 2024*

- **CGPA: 3.69/4.00 (Rank: 9/59)**
- **Notable Courses:** Machine Learning, AI, Human-Computer Interaction, Data Structures & Algorithms

Research Experience

Multilingual Sentiment Analysis of Social Media Reactions to OpenAI's Ghibli-Style Image Generation Feature
[Manuscript in preparation] — M.Sc. thesis; supervised by Dr. Israt Jahan

- Built a 36.7K-record multilingual corpus of Twitter and YouTube reactions to a viral generative-AI feature to study public sentiment toward AI-generated art, using machine translation of non-English posts, emoji-aware preprocessing, and VADER-based weak supervision.
- Benchmarked 21 transformer and hybrid Transformer-RNN configurations on both accuracy and calibration (ECE); a lightweight 1-layer DistilBERT-BiGRU hybrid nearly matched BERT-base in accuracy while achieving the lowest calibration error of all models.
- Found that accuracy varied by under 3 percentage points across models while calibration error varied more than fourfold, and that 16 of 18 hybrids were worse calibrated than their base transformer.

Deep Learning Based Bangladeshi Currency Coin Recognition
[Published at ICREST 2025, IEEE Xplore] — B.Sc. thesis; supervised by Md. Golam Moazzam

- Addressed coin identification for visually impaired users, as Bangladeshi coins are nearly identical in size and texture, by building a 14-class dataset of 2,800 smartphone images including augmented images.
- Trained a Faster R-CNN (ResNet-50 FPN) in PyTorch and compared it with a standard CNN baseline, with Faster R-CNN performing better across all metrics.

A Comparative Analysis of Low-Resource Techniques for Named Entity Recognition in Bengali
M.Sc. course research project (NLP)

- Investigated the effectiveness of modeling approaches for Bengali by benchmarking CRF, BiLSTM-CRF, mBERT, and XLM-RoBERTa on a custom 25K-sentence dataset merged from three public corpora.
- Enhanced models with FastText embeddings, character-level CNNs, and linguistic features; XLM-RoBERTa achieved the best token-level F1 (0.758), outperforming the classical CRF and BiLSTM-CRF baselines.

Cyberbullying Detection on Social Media Platforms Using Machine Learning
B.Sc. course research project (ML)

- Relabeled a 16K-comment Bengali toxic-comment dataset into 3 classes (Non-Bullying, Bullying, Crime) and compared four TF-IDF-based classifiers (LR, DT, RF, XGBoost); XGBoost with class-imbalance weighting led in accuracy, while Logistic Regression achieved the best F1.

Work Experience

Software Quality Assurance Engineer | Zelf AI Brooklyn, NY, USA | **Dec 2024 – Present**

- Perform root-cause analysis of bugs, audit production AI features (translation, sentiment, summarization, AI insights) via cloud logs, database queries, and API testing; reported 50+ previously undetected defects.
- Build and maintain automated test suites in Playwright alongside manual functional, regression, and UAT cycles; track issues in Jira, coordinating with developers and stakeholders.

Junior SQA Engineer | Dynamic Solution Innovators Dhaka, Bangladesh | **Apr 2024 – Dec 2024**

- Served as the sole QA engineer on the KeyPleaz project, a US real estate app; authored 500+ test cases in Qase and owned end-to-end testing of high-risk features including chat, calling, and Stripe payments.
- Presented demos to the US client and reviewed user stories with the PM, BA, and developers to keep requirements aligned with implementation.

Awards & Recognition

NST Fellowship | Ministry of Science and Technology: Awarded **$500** under the Physical Science category to support M.Sc. research. **2025 – 2026**

2nd Runner-up, Zelf Hackathon 2.0: Placed **3rd** among 300+ candidates in a QA-focused hackathon, leading to a recruitment interview and subsequent hire. **Oct 2024**

Merit Scholarship | Jahangirnagar University: Received **$100**/year during both B.Sc. and M.Sc. for academic excellence (Top 10%). **2019 – 2025**

Competitive Programming: Solved **700+** problems on LeetCode, Codeforces, and UVa; participated in contests including ICPC 2021 and ADA Lovelace NGPC 2021. **2019 – 2024**

Test Scores & Certifications

IELTS Academic: Overall 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0) **Jun 2026**

Fundamental Information Technology Engineer (FE) Examination – Full Passer **Oct 2023**

- Ranked **35th** nationally in the ITEE, a national-level IT engineers' examination governed by IPA, Japan.

Projects

OBE-Based Result Management System | *Python, Django, Bootstrap, SQLite* [GitHub]

- Built a role-based academic results platform where students view results, teachers analyze courses via auto-generated CO-PO attainment graphs, and administrators monitor performance department-wide.

Konta Niben – an E-commerce System | *MERN, Tailwind CSS, Jest, GitHub Actions* [GitHub]

- Developed a full-stack e-commerce platform following Agile/SDLC practices across 2 sprints, with JSDoc documentation, Jest unit tests, and CI via GitHub Actions.

IoT-Based Plant Watering System | *Arduino C++, NodeMCU ESP8266, Blynk, DHT11* [GitHub]

- Built a NodeMCU ESP8266 system that streams soil moisture, temperature, and humidity to the Blynk cloud, with phone alerts and remote pump control.

JU CSE Routinery | *Java, Android SDK, SQLite* [GitHub]

- Built an Android app that replaces static class-routine files with batch-wise daily and full schedules for students, plus direct class add/update for admins.

BoiPoka | *React, Vite, Tailwind CSS, Supabase, PostgreSQL* [GitHub]

- Built a social bookshelf app where readers track books with ratings, notes, and favorite quotes on public shelves they can browse each other's collections through.

Skills & Technologies

Languages: Python, C++, C, Java, JavaScript, SQL
ML/AI: PyTorch, scikit-learn, OpenCV, NLTK, Hugging Face Transformers, NumPy, Pandas, Matplotlib
Web Development: React, Django, Node.js, REST APIs, HTML5, CSS3, Bootstrap 5, Tailwind CSS
Testing & Automation: Playwright, Cypress, Postman, Qase
Databases: PostgreSQL, MySQL, MongoDB
Tools & Platforms: Git & GitHub, Linux, LaTeX, VS Code, Azure Boards, Jira/Trello, Slack

Extracurricular & Leadership Activities

Founding Member | JU Robotics Club **Feb 2023 – Oct 2024**

- Co-founded the club; built a 6-DoF robotic arm and line follower robots, and competed in inter-university events including Technocrats V.1 (IUBAT).

Publication Secretary | JU Computer Club **Jun 2021 – Apr 2024**

- Promoted from Executive Member to Publication Secretary within a year; managed the club's publication team for nearly 3 years, designing posters and promotional materials for events and recruitment drives.